Computer Science & IT



Fundamental of C Language

Data Types & Operator

Lecture No. 03



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Recap of Previous Lecture



- Topic — structure of program
- Topic — main function
- Topic — block, header file
- Topic — Declarative statement
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

C - language token

Data type

Slide



Topic : Primitive Data Types

- int
- character
- float

Integer :-

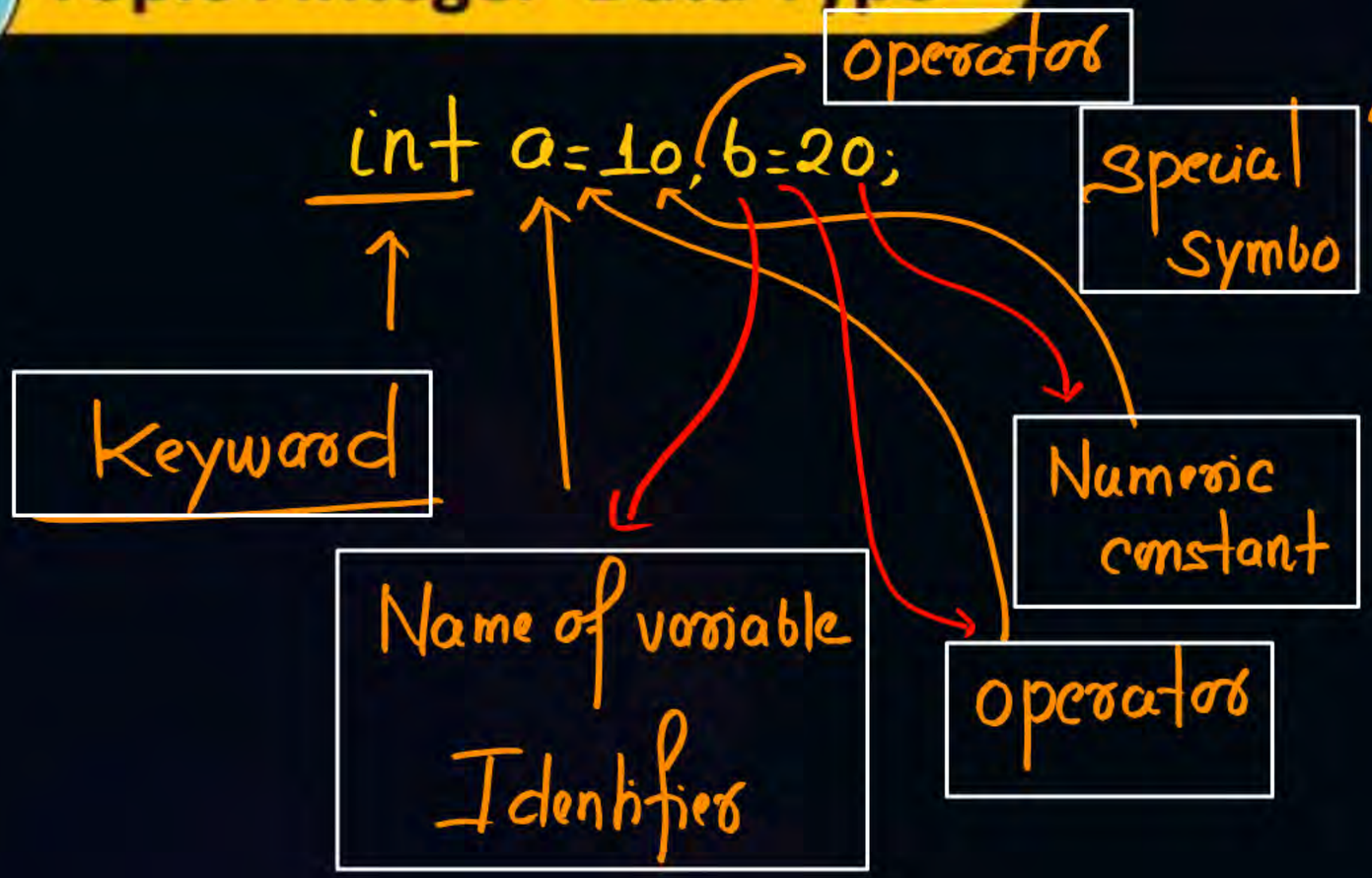
character :-

Real No. (floating point No)

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Topic : Integer Data Type



Token : Sequence of alphabets (words) that carries logical meaning to C. language.

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Keywords : Some words are reserved by Clanguage that can't be used called keywords

int , if, else, for, do, while ect.

Identifier : Identifiers are Name of a variable

Rule: Name of Identifier must start with alphabet or underscore followed by albabet, underscore & digit.

Identifiers

C - language case sensitive language

alphabets: a, b, c ... z
alphabets: A, B, C ... Z

underscore: a_b → underscore

digit: 0, 1, 2, 3 ... 9

Operator

1. Arithmetic operator
2. Relational operator
3. Logical operator
4. Bit wise operator

Conditional operator

Constant

```
int a = 10;   int b = -10;
```

integer decimal No.

binary constant : `int a = 0b101;`

octal constant : `int a = 015;`

Zero

Hexa decimal : `int a = 0xa1;`



Topic : Tokens

* Binay constant (0,1) starts with 0b followed by 0&21

* Octal constant (0,1,2,3,4,5,6,7) : starts with 0 followed

  0,1,2,3.4,5,6,7. in any order.

* Hexadecimal constant: start 0x (zero and x) followed

  0,1,2,3,4,5,6,7,8,9,a,b,c,d,e,f

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Topic : Unsigned Integer Representation

integer — int a;

data type

↓

Compiler allocates memory for integer a

- 2 Byte (DOS)
- 4 Byte (Windows)

4 Bytes = 4 X 8 = 32 bits



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Topic : Unsigned Integer Representation

Data will hold Limited value (Range of values)

integer comes modifier
- Signed
- unsigned

int a = 10; By default if No modifier then its a signed No

signed int a = 10;

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Topic : Unsigned Integer Representation

Singed a:10;

unsigned int a; a will take only posihve No.

Slide



Topic : Unsigned Integer Representation

unsigned int a = 10,

Size : 4B = 32 bits

Range of value

Slide



Topic : Signed Interger Representation

int a = -10;

It will take +ve & -ve No.

signed int a = -10;

Slide



Topic : Signed Interger Representation

int a = -10;

Signed int a = -10;

It will take +ve & -ve No.



Negative No. representation

2's complement form

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THANK - YOU

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